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Equine nasogastric intubation adapter with backflow diverter

Abstract

A rigid hollow tubular adapter releasably attached at a longitudinal nozzle end to a flexible tube during equine nasogastric intubation comprising an annular sleeve having a stationary tubular member and a slidable tubular member. The slidable tubular member is delimited at its ends by the stationary tubular member and a detent member disposed between an equine reflux drainage hole and a longitudinal mouthpiece end. Operator's puffs of airflow through a mouthpiece urges the slidable member to slide toward the detent member closing the equine drainage hole to enable passage of medication into the equine stomach. Equine reflux urges the slidable tubular member to slide toward the stationary tubular member opening the reflux drainage hole to divert equine backflow downwardly through the equine backflow drainage hole preventing passage of equine reflux into the operator's mouth.

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Background/Summary

RELATED APPLICATIONS (1) This application is a continuation-in-part of pending U.S. application Ser. No. 16/570,479 filed Sep. 13, 2019 which is a continuation-in-part of U.S. application Ser. No. 15/045,232 filed Feb. 16, 2016 now abandoned.

FIELD OF THE INVENTION

(1) The present invention generally relates to nasogastric intubation, and more particularly relates to a portable apparatus enabling an operator, typically a veterinarian or other medical professional, to safely conduct equine nasogastric intubation without jeopardizing the horse's well-being and simultaneously avoiding ingestion or aspiration of contaminated air emanating from breaths during equine coughing and also avoiding ingestion of contaminated liquid emanating from an equine stomach into the operator's mouth while nasogastric intubation is being conducted.

BACKGROUND OF THE INVENTION

(2) As is well known by veterinarians specializing in equine practice, a horse is a non-ruminant herbivore having a digestive system configured with unique anatomical attributes. For instance, a horse has a large gastrointestinal tract about one hundred feet long, primarily consisting of the small and large intestines. The small intestine interconnects the stomach with the large intestine, through its two inch diameter and constitutes about 30% of the equine digestive tract. It is typically fifty to seventy feet long having a capacity of about ten to fifteen gallons. Most food is digested and absorbed into the bloodstream in the small intestine, with any residual roughage and liquids being transferred to the large intestine.

(3) The equine large intestine is typically fifteen to twenty feet long having a diameter as small as two to three inches in one portion, to eight to ten inches in another portion, and as large as about twenty inches in another, lower abdominal portion thereof. Thus, the equine large intestine has

diameters throughout that are significantly larger than the diameter of the small intestine—with a capacity of more than thirty gallons. Bacteria residing therein enable digestion of cellulose plant fiber via fermentation, wherein plant fiber is broken down into carbohydrates which may then be readily absorbed in the large intestine. A horse has a relatively small, single-compartment stomach typically having a capacity of about 4 gallons, corresponding to about ten percent of the digestive system. Unlike other species, the oral sphincter of the equine stomach is very tight and doesn't allow retrograde movement of any material. Accordingly, horses are unable to vomit or eructate gas.

(4) Digestion commences in the oral cavity when a horse ingests plant food. The ingested food is masticated in the mouth and then transported to the stomach through an elongated esophagus tube, typically four to five feet long, urged by peristaltic wave-like contractions that are propagated by a well-developed muscular ring interconnecting the esophagus therewith.

(5) As is also well known by practitioners in the art, colic manifest as abdominal pain constitutes a common equine emergency attributable to such circumstances as attenuated movement of the small intestine and/or obstructions therein; inhibited movement of the large intestine and/or obstructions therein and/or twisting or displacement thereof. The result of these phenomena could include devolve to accumulation of fluid in a horse's stomach. A known factor causing such problematic accumulation is a horse being unable to eructate. The equine cardiac sphincter is so tight wherein passage of gas, liquid or other material retrograde is disallowed. Thus, it will be appreciated there are several underlying causes for colic which, of course, range from mild and inconsequential to severe and life-threatening. Accordingly, in the abundance of caution, every occurrence of colic should preferably be handled with urgency, as the equine stomach or intestine can rupture due to backup pressures during gastrointestinal disfunction.

(6) Nasogastric intubation is routinely invoked by veterinarians to treat equine medical conditions such as colic and chokes, or to gain access to the stomach for evaluation or to deliver medication. In its most common application, nasogastric intubation is used during colic treatment to facilitate decompression of the stomach and to administer therapy. For equine chokes, nasogastric intubation is used to clear occlusions of the esophagus. It should be understood that such obstructions must be promptly remedied to avoid feed and fluid being urged to pass into an occluded oropharynx forcing these materials toward the lungs as a horse attempts to swallow. As is known in the art, notwithstanding advent of paste dewormers to clear occlusions in the esophagus, parasiticides are still routinely administered by invoking a nasogastric tube (herein abbreviated “NGT”) to expedite direct access to the troubled equine stomach. And, finally, practitioners in the art will also readily recognize that a NGT is also especially useful for enabling gastric endoscopy to apprise a medical professional of the integrity and condition of equine stomach walls and contents therein.

(7) Once a NGT is emplaced into the ventral nasal passage and into the equine oropharynx the stimulus initiates swallow reflex forcing the NGT into the esophagus. Then a veterinarian can blow small puffs of air into and thereby dilating the equine esophagus as the NGT is passed into an esophagus simultaneously facilitating the NGT's passage. This intermittent blowing via the NGT tends to facilitate dilation of an otherwise collapsed equine esophagus, thereby allowing contemplated aboral movement and subsequent insertion of the NGT into the equine stomach. It will be appreciated by those conversant in the art the NGT would be advanced until essentially being situated level with the equine twelfth rib.

(8) Upon entry into the equine esophagus, after orally effectuating prerequisite vacuum by imparting a modicum of suction into the NGT, a veterinarian can confirm the NGT has been emplaced properly in the equine esophagus rather than the equine trachea—if and when the veterinarian hears gas and/or smells equine stomach contents. It should be appreciated there would be an absence of negative pressure generated from this modicum of suction, if the NGT were incorrectly emplaced into the trachea. Without this functionality, a veterinarian would be left auscultating gas or smelling gastric contents which may or may neither be heard nor present.

(9) It should be appreciated by those conversant in the art absent backflow preventive apparatus functioning as a check valve as herein disclosed, stimulating or tickling the equine pharynx frequently would effectuate aspiration of equine stomach contents into a veterinarian's lungs. Furthermore, since there would typically be a formidable probability of spontaneous equine reflux occurring, a veterinarian would be exposed to pungent, fermented or even bloody reflux constituting a health hazard and a biohazard if a backflow preventive apparatus were not invoked.

(10) The prior art appears to be devoid of any convenient and sufficiently portable apparatus that reliably enables a veterinarian or other medical professional to safely engage in equine nasogastric intubation as contemplated herein.

SUMMARY OF THE INVENTION

(11) The present invention contemplates an equine nasogastric intubation adapter structured to allow an operator, typically a veterinarian or other medical professional, to manually administer suction or puffs of air while simultaneously preventing or diverting equine stomach reflux from entering the operator's mouth. More particularly, embodiments of the present invention devolve to an adapter facilitating an operator effectuating safe passage of a NGT upwardly through the equine ventral nares into the nasopharynx and pharynx, wherein the horse is induced to swallow thereby urging the NGT into the equine esophagus.

(12) It will be readily understood the equine esophagus is typically collapsed in its resting state. So, prior to conducting nasogastric intubation, an operator must cause this equine esophagus to be expanded to permit and accommodate NGT-passage therethrough. Hence, the operator would preferably blow gently into an adapter embodiment wherein air would travel therethrough into the NGT to apply minimal pressure to urge the equine esophagus to become dilated.

(13) Embodiments of the instant portable equine nasogastric longitudinal intubation adapter have been configured to be easily and conveniently attached at a female end of a variety of suitably sized flexible tubes or flexible rubber hoses. It will be seen such embodiments comprise a lightweight rigid hollow tubular member having an annular sleeve member contained axially therein. The annular longitudinal sleeve member comprises a first stationary member configured to be affixed to one end of the tubular adapter and a second slidable tubular member configured to be slidably displaced from being disposed adjacent the stationary tubular member to being disposed remotely thereof as will be hereinafter described. Thus, the slidable tubular member is readily urged to slide within the annular sleeve member from a first position adjacent the stationary tubular member to a second position remote of the stationary tubular member toward the opposite end of the tubular adapter until abutting and adjacent a detent member. It will be evident to those skilled in the art each of these two delimiting positions is disposed on opposite sides of an equine reflux drainage hole which is, in turn, disposed between the stationary tubular member and the detent member.

(14) The proximal end of the nasogastric intubation adapter has a mouthpiece to be emplaced within the operator's mouth simultaneously with the other distal, opposite end thereof having a nozzle member preferably circumscribed by a series of undulating diminishing diameters or ridges; or a tapered threaded screw configuration, conveniently and securably releasably attached to a flexible tube or flexible rubber hose sized to comfortably and safely be inserted into the equine digestive system through the equine nasal cavity. Preferred NGT adapter embodiments comprise a mouthpiece having a pair of bite-tabs with a pliable plastic shield enabling hands-free gripping and, more particularly, teeth-gripping thereof at the proximal end to sustain its uninterrupted presence in the operator's mouth while appropriate treatment is being administered.

(15) It is an object and advantage of embodiments of the present invention to serve as a safety interface while providing a convenient, inexpensive and lightweight portable adapter for attaching to a flexible tube or flexible hose incorporated into an equine nasogastric intubation protocol.

(16) It is another object of the present invention to provide a disposable equine nasogastric intubation backflow diverter adapter tube that minimizes risk to an operator attributable to equine contamination and concomitant biohazards.

(17) It is yet another object of the present invention to provide a disposable equine nasogastric intubation backflow diverter adapter tube for facilitating safe and convenient emplacement of a nasogastric tube into an equine stomach.

(18) These and other objects and advantages of the present invention will become apparent from the following specification and accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 depicts a perspective side view from the mouthpiece end of an embodiment of the present invention, with the backflow drainage hole disposed in an open position.

(2) FIG. 2 depicts a side view of an embodiment of the present invention, with the backflow drainage hole disposed in an open position.

(3) FIG. 3 depicts a side view of the embodiment depicted in FIG. 1, with the backflow drainage hole disposed in a partially-open position.

(4) FIG. 4 depicts a side view of the embodiment depicted in FIG. 1, with the backflow hole disposed in a closed position.

(5) FIG. 5 depicts an opposite side view of the embodiment depicted in FIG. 4.

(6) FIG. 6A depicts an end view of the embodiment depicted in FIG. 1.

(7) FIG. 6B depicts the opposite end view of the end view depicted in FIG. 6A.

(8) FIG. 7 depicts a top view of the embodiment depicted in FIG. 1.

(9) FIG. 8 depicts a bottom view of the embodiment depicted in FIG. 1.

(10) FIG. 9 depicts a perspective side view from the mouthpiece end of another embodiment of the present invention, with the backflow hole disposed in an open position.

(11) FIG. 10 depicts a top view of the embodiment depicted in FIG. 9.

(12) FIG. 11 depicts a bottom view of the embodiment depicted in FIG. 9, with the backflow hole disposed in an open position.

(13) FIG. 12 depicts a bottom view of the embodiment depicted in FIG. 9, with the backflow hole disposed in a partially-open position.

(14) FIG. 13 depicts a bottom view of the embodiment depicted in FIG. 9, with the backflow hole disposed in a closed position.

(15) FIG. 14A depicts an end view of the embodiment depicted in FIG. 9.

(16) FIG. 14B depicts the opposite end view of the end view depicted in FIG. 14A.

(17) FIG. 15 depicts a side view of the embodiment depicted in FIG. 9.

(18) FIG. 16 depicts the opposite side view of the embodiment depicted in FIG. 15.

(19) FIG. 17 depicts a side view of a streamlined embodiment of the present invention, with the backflow drainage hole disposed in an open position.

(20) FIG. 17A depicts a cross-sectional view of the embodiment depicted in FIG. 17.

(21) FIG. 18 depicts a top view of the embodiment of the present invention depicted in FIG. 17, with the backflow drainage hole disposed in a closed position.

(22) FIG. 19 depicts a bottom view of the embodiment depicted in FIG. 18.

(23) FIG. 20 depicts an end view of the embodiment depicted in FIG. 17.

(24) FIG. 21 depicts the opposite end view depicted in FIG. 20.

DETAILED DESCRIPTION

(25) Reference is made herein to the figures in the accompanying drawings in which like numerals refer to like components. Referring collectively to FIGS. 1, 2-5, 6A-6B, 7-8, 9, 10-13, 14A-14B, and 15-16, there are depicted various views of two alternative embodiments of the present invention. One embodiment of equine nasogastric intubation backflow diverter adapter 2 is depicted in perspective view in FIG. 1 and another backflow diverter adapter embodiment 5 is

depicted in perspective view in FIG. 9. Yet another, more streamlined embodiment of equine nasogastric intubation backflow diverter adapter 7 is depicted collectively in various views in FIGS. 17-17A, 18-21.

(26) More particularly, referring collectively to FIGS. 1-8, there is depicted equine nasogastric hollow tubular backflow preventer adapter embodiment 2 having substantially flat external member 15A fixedly disposed atop exterior surface 20 of rigid hollow tubular member 10, and disposed axially and centrally therewithin as shown. Exterior surface 20 is preferably configured with beveled or chamfered edge 12 to enhance interconnection with a mouthpiece member 65 disposed at end 22B of rigid hollow tubular member 10 as will be hereinafter described. Annular tubular sleeve member 40 having a circular cross-section is disposed longitudinally upon interior surface 30 of rigid hollow tubular member 10. Bore 35 is circumscribed by interior surface 30 and disposed both axially and centrally therethrough.

(27) Disposed oppositely of end 22B, end 22A of rigid hollow tubular member 10 is configured with nozzle member 60 preferably circumscribed by series of undulating diminishing diameters or ridges 62 as the nozzle remote end is approached. Thus, nozzle member 60 is configured to be releasably and securably attached to a flexible tube or flexible hose T sized to be comfortably and safely inserted into an equine digestive system through an equine nasal cavity (not shown). It will be appreciated that end 22B of rigid hollow tubular member 10 is preferably configured with snorkel-like mouthpiece 65 having pair of bite-tabs 70 with a pliable plastic shield enabling an operator's hands-free, mouth-teeth-gripping thereof at the proximal end to sustain its uninterrupted presence in the mouth of an operator administering the nasogastric procedure contemplated hereunder.

(28) Still focusing on FIGS. 1-5, 6A-B, and 7-8, there are seen different views of the backflow diverter aspect contemplated by the present invention. Thus, annular tubular sleeve member 40 disposed axially and centrally within rigid hollow tubular member 10 comprises a first, stationary tubular member 45 and a second, slidable tubular member 50, configured to be disposed adjacent each other and preferably being congruent, having substantially the same diameter and material of construction.

(29) Stationary member 45 and slidable member 50 are disposed in an adjacent relationship, when there is equine stomach backflow or when the horse coughs thereby instigating movement from nozzle end 22A toward mouthpiece end 22B, in turn, urging slidable member 50 to slide toward stationary member 45 wherein end 47B of stationary member 45 abuts proximal end 52B of slidable member 50. Thus, under this equine backflow condition, positioning of slidable tubular member 50 within axial bore 35 is delimited by corresponding end member 47B of stationary tubular member 45 and end member 52B of slidable tubular member 50, respectively, wherein backflow drainage hole 25 is disposed in an open position (See FIGS. 1 and 2).

(30) Contrariwise, to commence the nasogastric intubation procedure contemplated hereunder, an operator would impart air flow through mouthpiece end 22B of rigid hollow tubular member 10 by gently blowing into the mouthpiece 65 causing the equine esophagus to expand. It will be appreciated that this gentle airflow travels around sliding member 50 thereby urging it to slide toward and cease movement at detent 55 thereby completely covering equine drainage hole 25 (see FIG. 4) enabling uninterrupted, continuous flow of medication or the like into the equine stomach as intended. Once proper placement of NGT T has been accomplished, embodiments 10 of the instant equine nasogastric intubation adapter would be removed.

(31) Those skilled in the art will appreciate that equine backflow would engender greater momentum than gentle blows of an operator's breath so that positioning of slidable member 50 would be urged to slidably move from being emplaced adjacent detent member 55 at end member 22A of rigid hollow tubular member 10 to being emplaced adjacent end member 47B of stationary member 45 at end member 22B of rigid hollow tubular member 10, assuring opening of equine reflux backflow hole 25.

- (32) It will be understood that operator-generated gentle airflow through mouthpiece **65** urges separation of slidable member **50** from stationary tubular member **45** through rigid hollow tubular member **10** until positioning of slidable member **50** within bore **35** is delimited by detent member **55**. Accordingly, slidable member **50** abuts detent member **55** at edge **52A** thereof, assuring closure of reflux backflow drainage hole **25** (See FIGS. 4-5).
- (33) It will also be understood by those skilled in the art, that, as depicted in FIG. 3, the position of slidable member **50** within bore **35** has been slidably urged by equine backflow to the extent of only being partially moved toward detent member **55**, wherein equine reflux backflow drainage hole **25** is emplaced essentially in a “half-open” position. It will be seen that detent member **55** and stationary member **45** are disposed on opposite sides of backflow drainage hole **25**, wherein detent member **55** comprises a firm nipple member affixed to and protruding minimally from interior surface **30** of rigid hollow tubular member **10**. To afford sufficient detent capability for effectively stopping further sliding progress of slidable tubular member **50**, the nipple member should be constructed from solid material to be firm enough to cease movement of slidable tubular member **50** toward the longitudinal nozzle end of rigid hollow tubular member **40** as contemplated hereunder, wherein equine drainage hole **25** would be completely covered. For instance, such nipple member could constitute a simple cut-off nail-head or similar cut-off screw-top affording sufficient height differential above interior surface **30** to prevent further sliding movement of slidable tubular member **50**—beyond this prescribed sliding-movement-limiting position as depicted herein.
- (34) As will become evident to those skilled in the art, the present invention contemplates that equine backflow drainage hole **25** would be open when equine liquid reflux or like effluent effectuates backflow pressure in NGT T attached to end **22A** of adapter **2** to urge slidable member **50** to slide from emplaced adjacent detent member **55** toward opposite end **22B** of rigid hollow tubular member **10** of adapter **2** to another position adjacent stationary member **45**. When common end **52B** of slidable member **50** is emplaced adjacent of and delimited by common end **47B** of stationary member **45**, its exterior surface **20** is clear of backflow drainage hole **25**, thereby enabling backflow drainage hole **25** to be disposed in an open position enabling equine backflow to be diverted downwardly through equine backflow drainage hole **25** wherein equine backflow is inherently prevented from entering an operator's mouth.
- (35) Mouthpiece member **65** is releasably attached to end **22B**, disposed oppositely of end **22A** of rigid hollow tubular member **10**, to enable adapter embodiments hereof to be securely inserted into an operator's mouth and sustained therein hands-free relying upon his teeth biting into pair of bite tabs **70** during the equine nasogastric intubation procedure. In this arrangement, as elucidated hereunder, equine reflux is diverted from remaining in the main path from the equine stomach into the operator's mouth, and, instead, urged to flow into a fail-safe diverted downward path through drainage hole **25** into a reflux collection pail or an otherwise designated repository such as the ground (not shown).
- (36) It will also be appreciated by those conversant in the art that another embodiment of the present invention could be similarly configured with a single diameter tubular member and with a loop member rather than a flat member. More particularly, FIGS. 9-13, 14A-14B, and 15-16 depict another embodiment of equine NGT backflow preventer adapter **5** having an external member **15B** fixedly disposed atop external surface **20** of rigid hollow tube member **10**. External surface **20** is preferably configured with beveled or chamfered edge **12**. Annular sleeve **40** having circular cross-section is disposed upon interior surface **30** of rigid hollow tubular member **10** as has been hereinbefore described. Bore **35** is circumscribed by interior surface **30** and both axially and centrally disposed therethrough.
- (37) Focusing collectively on FIGS. 9-13, 14A-B, and 15-16, there are seen different views of the backflow diverter aspect taught by embodiments of the present invention. Thus, interior surface **30** fixedly receives stationary member **45** and slidable member **50**, configured to be disposed adjacent

each other and preferably congruent and having the same material of construction. When disposed in an adjacent relationship as depicted in FIG. 11, positioning of slidable member 50 within bore 35 is delimited by corresponding common end 47B of stationary member 45 and common end 52B of slidable member 50, respectively, wherein reflux backflow drainage hole 25 is disposed in an open position. Contrariwise, when slidable member 50 is slidably separated from stationary member 45, as depicted in FIGS. 12 and 13, positioning of slidable member 50 within bore 35 is delimited by detent member 55, wherein backflow drainage hole 25 is disposed in a closed position. Referring again collectively to FIGS. 8-14, it will be seen that detent member 55 and stationary member 45 are disposed on opposite sides of equine backflow drainage hole 25 (see end 47B of stationary member 45 and detent member 55 disposed oppositely of drainage hole 25).

(38) It will also be appreciated by those conversant in the art that yet another embodiment of the present invention could be similarly configured with a single diameter tubular member featuring a more streamlined and even lighter weight linear structure absent either an external loop member or absent an external flat member. More particularly, FIGS. 17-17A, and 18-21 depict such a streamlined embodiment of equine NGT backflow preventer adapter 7 having a streamlined external surface 20 of rigid hollow tubular member 10'. It should be understood FIGS. 17-17A, and 18-21 depict the progression and interrelationship between annular tubular sleeve member 40 disposed longitudinally upon interior surface 30' of rigid hollow tubular member 10 (see adapter embodiment 2 depicted in FIGS. 2-4) and annular tubular sleeve member 40' disposed longitudinally upon interior surface 30 of rigid hollow tubular member 10' (see adapter embodiment 7 depicted in FIGS. 17-17A, and 18-19), respectively.

(39) More particularly, FIG. 17 depicts a side view of a streamlined embodiment of the present invention, with backflow drainage hole 25 disposed in an open position. FIG. 17A depicts a cross-sectional view of the embodiment depicted in FIG. 17 identifying the communication between first primary axial bore 35 and secondary smaller axial bore 36—relative to longitudinal channel member 37 and slidable member 50 and end portion thereof 52A; showing open, unobstructed equine backflow fluid drainage hole 25. As shown, when end member 52A of slidable member 50 is positioned before drainage hole 25, the drainage hole is open for receiving backflow of equine air and/or backflow of equine stomach fluids. FIG. 18 depicts a top view of the embodiment depicted in FIG. 17, with backflow drainage hole 25 disposed in a closed position; and FIG. 19 depicts a bottom view corresponding to the top view thereof depicted in FIG. 18, with the backflow hole disposed in a closed position evidenced by edge member 52A of slidable member 50 being stopped at detent member 55 and simultaneously covering drainage hole 25. FIG. 20 depicts an end view of the embodiment depicted in FIG. 17; and FIG. 21 depicts the opposite end view of the end view depicted in FIG. 20.

(40) External surface 20 is preferably configured with beveled or chamfered edge 12. Annular sleeve 40' having circular cross-section is disposed upon interior surface 30' of rigid hollow tubular member 10' as has been hereinbefore described. Bore 35 is circumscribed by interior surface 30' and both axially and centrally disposed therethrough. Unlike hereinbefore described embodiments 2 and 5 of the equine nasogastric tube backflow diverter adapter contemplated hereunder, embodiment 7 has been constructed with a streamlined configuration having longitudinal channel 37 disposed axially of rigid hollow equine nasogastric tubular backflow diverter adapter depicted in FIGS. 17-17A, and 18-19 for receiving the operator's gentle puffs of air via mouthpiece member 65' and communicating this airflow through secondary smaller axial bore 36 into nozzle member end 60 to cause the equine esophagus to dilate as hereinbefore described. It will also be understood slidable tubular member 50 is caused to slide toward and cease movement at detent member 55, thereby completely covering equine drainage hole 25 enabling uninterrupted continuous flow of medication or the like into the equine stomach.

(41) It has been found sufficient airflow is manifest to achieve the airflow purposes herein elucidated when longitudinal channel 37 has been cut into interior surface 30' with a depth of

substantially 40-50% of the tubular wall thickness delineated by different thickness of exterior tubular surface **20** compared with interior tubular surface thereof **30'**. As shown by the arrowheads contained within mouthpiece **65'** in FIGS. **18** and **19**, respectively, while NGT adapter **7** is secured in the operator's mouth hands-free by the operator biting firmly on pair of bite-tabs **70'** gentle puffs of air travel into longitudinal channel **37** with entry thereinto at edge **47A** and flow around solid member **50** and then through secondary axial bore **36** nozzle member **60'** (see arrow heads depicting airflow from mouthpiece end to nozzle end of rigid hollow tubular member **10'**).

(42) Now focusing collectively on FIGS. **17-17A**, **18-19** and **21**, there is seen advantageous application of preferably internally-threaded rubber grommet **80** for easily and reliably securing attachment and detachment of nozzle member **60'** to flexible nasogastric hose T. It should be appreciated by those skilled in the art the pitch and pitch diameter manifest on tapered screw configuration **62'** of nozzle member **60'** have been proportionately increased and widened compared to the series of diminishing diameters of nozzle members **62** configured on adapter embodiments **2** and **5** to facilitate improved temporary attachment of nozzle member **60'** to nasogastric hose T. Now focusing collectively on FIGS. **17-17A** and **18-20**, there is seen a streamlined configuration manifest on mouthpiece **65'** for embodiment **7** which is devoid of the ridge between lip and teeth depicted in FIGS. **2-4** for embodiment **2** and depicted in FIG. **15** for embodiment **5**.

(43) Such improved attachment has been found to be advantageous since hose T when in situ would typically be warm and soft rendering secure attachment and detachment therewith to be elusive. When a horse is being treated as contemplated hereunder it may become necessary and awkward to quickly detach the nasogastric tube T from nozzle member **60'** or, alternatively, to expeditiously attach the nasogastric rubber tube T to nozzle member. Accordingly, including firm rubber grommet **80** integral to this attachment-detachment protocol significantly improves not only the efficacy of the connection therebetween, but also expedites achieving such attachment and detachment, per se. To exemplify typical grommet components, but not intended to limit grommets' materials of construction or the like, Jefferson Rubber Works, Inc. of Worchester, Massachusetts is a source of suitable rubber grommets contemplated for application hereunder. It will appreciated grommet member **80** should preferably be constructed with internal threads to enable it to be securely establish contact with, i.e., effectively "bite" into or "grab" upon, interior surface of rubber hose T. Then to tighten and thus secure the connection between the grommet and the rubber hose, the grommet would be rotated in a clockwise direction. Alternatively, to loosen the connection between the grommet and the rubber hose, the grommet would be rotated in a counterclockwise direction, thereby enabling expeditious detachment thereof.

(44) As hereinbefore elucidated, interior surface **30'** fixedly receives stationary member **45** and slidable member **50**, configured to be disposed adjacent each other and preferably congruent and having the same material of construction. When disposed in an adjacent relationship, positioning of slidable member **50** within bore **35** is delimited by corresponding common end **47B** of stationary member **45** and common end **52B** of slidable member **50**, respectively, wherein reflux backflow drainage hole **25** is disposed in an open position. Contrariwise, when slidable member **50** is slidably separated from stationary member **45**, positioning of slidable member **50** within bore **35** is delimited by detent member **55**, wherein backflow drainage hole **25** is disposed in a closed position. Referring again collectively to FIGS. **17-17A**, **18-21**, it will be seen that detent member **55** and stationary member **45** are disposed on opposite sides of equine backflow drainage hole **25** (see end **47B** of stationary member **45** and detent member **55** disposed oppositely of drainage hole **25**).

(45) The following is a tabulation of the components depicted in the drawings:

(46) TABLE-US-00001 Component Listing Numeral Component Explanation
2 Equine nasogastric tube backflow Flat grip embodiment diverter adapter
5 Equine nasogastric tube backflow Loop grip embodiment diverter adapter
7 Equine nasogastric tube backflow Streamline

embodiment diverter adapter 10 Rigid Hollow Tubular Member 10' Rigid Hollow Streamlined Tubular Member 12 Beveled or chamfered edge 15A Flat member Rubberized 15B Loop member 20 Exterior surface 22A, B Respective ends of Rigid Hollow 22A: nozzle end Tubular Member 10 22B: mouthpiece end 22A', Respective ends of Rigid Hollow 22A': nozzle end 22B' Tubular Member 10' 22B': mouthpiece end 25 Backflow Drainage Hole, Lateral 30 Interior surface of Rigid Hollow Tubular Member 10 30' Interior surface of Rigid Hollow Tubular Member 10' 35 Primary Axial bore passing For NGT adapter embodiments 2 or 5, axial through primary rigid tubular bore 35 passes through primary rigid tubular member 10 or 10' member 10; for streamlined NGT adapter embodiment 7, axial bore 35 passes through primary rigid tubular member 10' 36 Secondary Axial bore passing For NGT adapter embodiment 7, axial bore 36 passes through primary rigid tubular through nozzle member 60 of rigid member 10' from mouthpiece end (65') tubular member 10' through members 40, 45, and 50 to nozzle end 60' 37 Longitudinal Channel Proximal to & cut into Interior surface 30'; thickness of Channel approximately same as distance between exterior wall 20 and interior wall 30' 40 Annular sleeve of Rigid Hollow Tubular Member 10 40' Annular sleeve of Rigid Hollow Tubular Member 10' 45 Stationary member of Rigid Hollow Tubular Member 10 47A, B Respective ends of Stationary End 47B receives end 52B of slidable member 10 member 50 50 Slidable member of Rigid Hollow Tubular Member 10 52A, B Respective ends of Slidable Member End 52B receives end 47B of stationary 50 member 45 55 Detent member Limiting movement of slidable member; nipple-like structure attached to and protruding minimally from interior surface 60 Nozzle member of Equine Receiving NGT nasogastric tube backflow diverter adapter 2 & 5 60' Nozzle member of Equine Receiving NGT nasogastric tube backflow diverter adapter 7 62 Series of diminishing diameters or Facilitate secure interconnection of flexible ridges, diminishing as remote, open tube or hose with adapter end of nozzle is approached 62' Tapered threaded screw configuration Facilitate secure interconnection of flexible tube or hose with adapter complemented by internally-threaded grommet 65 Mouthpiece member of Equine Disposed oppositely of nozzle nasogastric tube backflow diverter adapter 2 & 5 65' Mouthpiece member of Equine Streamlined configuration devoid of ridge nasogastric tube backflow diverter between operator's teeth and lip. adapter 7 Disposed oppositely of nozzle 70 Bite-Tab member of Equine Conveniently hold adapter in mouth, hands- nasogastric tube backflow diverter free adapter 2 & 5 70' Bite-Tab member of Equine Conveniently hold adapter in mouth, hands- nasogastric tube backflow diverter free adapter 7 80 Grommet Firm rubber having internal threads for gripping tube to nozzle member; rotating grommet in one direction tightens grip while rotating grommet in opposite other direction loosens grip T Flexible tube For nasogastric insertion purposes

(47) Other variations and modifications will, of course, become apparent from a consideration of the structures and techniques hereinbefore described and depicted. Accordingly, it should be clearly understood that the present invention is not intended to be limited by the particular features and structures hereinbefore described and depicted in the accompanying drawings, but that the present invention is to be measured by the scope of the appended claims.

Claims

1. A portable equine nasogastric intubation adapter comprising: a rigid hollow tubular member having at a first longitudinal end member and having a nozzle member disposed axially of said first longitudinal end for releasably being attached to a flexible tube or a flexible hose by an operator, while said operator positions an opposite second longitudinal end of said rigid hollow tubular member into said operator's mouth; an annular sleeve member disposed axially within said rigid hollow tubular member and circumscribing a first, primary axial centrally disposed bore and, in turn, said first, primary axial centrally disposed bore circumscribing a secondary axial centrally disposed bore; said annular sleeve member comprising a stationary tubular member affixed at said

first longitudinal end of said rigid hollow tubular member; and a slidable tubular member slidably disposed within said rigid hollow tubular member; a longitudinal channel member disposed parallel to said rigid hollow tubular member and cut into an interior surface of said rigid hollow tubular member having a depth of 40% to 50% of a depth between an exterior surface and said interior surface of said rigid hollow tubular member for communicating airflow from said operator's mouth through said secondary axial bore to an equine stomach; an equine reflux drainage hole disposed upon an interior surface of said rigid hollow tubular member and further disposed between said first longitudinal end and said second longitudinal end of said rigid hollow tubular member; said slidable tubular member urged to slidably move upon said rigid hollow tubular member between a first position adjacent said stationary tubular member and a second position toward said first longitudinal end of said rigid hollow tubular member abutting a detent member affixed upon and protruding upwardly from said interior surface of said rigid hollow tubular member and disposed between said equine reflux drainage hole and said first longitudinal end of said rigid hollow tubular member.

2. The portable equine nasogastric intubation adapter recited in claim 1, wherein said detent member comprises a firm nipple member.

3. The portable equine nasogastric intubation adapter recited in claim 1, wherein said nozzle member comprises a tapered threaded screw configuration circumferentially emplaced upon said nozzle member for conveniently and securably releasably attaching said flexible tube or flexible hose sized to be comfortably and safely inserted into an equine nasal cavity and passed therethrough to an equine esophagus and into an equine stomach.

4. The portable equine nasogastric intubation adapter recited in claim 3, wherein said nozzle member further comprises an internally threaded grommet member screwably received by said tapered threaded screw configuration.

5. The portable equine nasogastric intubation adapter recited in claim 4, wherein a connection between said nozzle member and said flexible tube or flexible hose is tightened when said internally threaded grommet member is rotated clockwise around said nozzle member and wherein said connection between said nozzle member and said flexible tube or flexible hose is loosened when said internally threaded grommet member is rotated counter-clockwise around said nozzle member.

6. The portable equine nasogastric intubation adapter recited in claim 1, wherein said opposite second longitudinal end of said rigid hollow tubular member comprises a snorkel-like mouthpiece member having a pair of bite-tabs with a pliable plastic shield disposed on each side of said mouthpiece member enabling said operator's hands-free, mouth-teeth gripping of said equine nasogastric intubation adapter while being inserted into said equine stomach through an equine nasal cavity.

7. The portable equine nasogastric intubation adapter recited in claim 6, wherein said equine reflux drainage hole is disposed in a closed position when said slidable tubular member is urged by gentle puffs of breath emanating from said operator's mouth into said mouthpiece member and through said rigid hollow tubular member to be slidably positioned toward and adjacent said detent member thereby covering said equine reflux drainage hole to enable passage of medication into said equine stomach.

8. The portable equine nasogastric intubation adapter recited in claim 7, wherein said equine reflux drainage hole is disposed in an open position when said slidable tubular member is urged by backflow of equine reflux entering into said rigid hollow tubular member through said nozzle member disposed at said first longitudinal end of said rigid hollow tubular member to slide away from said detent member toward said stationary tubular member until positioned adjacent thereto at said opposite second longitudinal end of said rigid hollow tubular member, thereby uncovering said equine reflux drainage hole to divert passage of said equine reflux downwardly through said equine reflux drainage hole and simultaneously disable passage of said equine reflux into said operator's

mouth.

9. The portable equine nasogastric intubation adapter recited in claim 1, wherein end members of said rigid hollow tubular member each comprise a bevel member.
